Arvan Kadkol

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EDUCATION

Homestead High School Class of 2028
GPA: 4.0 weighted/ 3.8 unweighted
Relevant Courses: AP Computer Science, Honors Precalculus, AP Physics 1
Clubs: Robotics (Scout/Data Collection Lead), Shogi, Computer Science
Project Portfolio: https://github.com/NotAPokemon

VOLUNTEER WORK

Volunteer | West Valley Elementary School **2018 - Present**

- Managed the flow of students through carnival games, ensuring activities ran efficiently and safely.
- Operated the concession stand, handling food distribution and coordinating with other volunteers.

EXTRACURRICULAR ACTIVITIES

Robotics Scout and Data Collection Lead | Homestead High School **2024 - Present**

- Dedicated 30+ hours per week to preparing for robotics competitions to improve team performance.
- Led a team of 6 students to design and develop a data-collection mobile app to support competition strategy.
- Reorganized the team's software architecture and built internal libraries, significantly improving development efficiency.

FBLA Mobile App Development Team | Homestead High School **2024 - 2025**

- Placed 2nd at the State competition, advancing to the National competition.
- Planned development timelines and milestones over a two-month development cycle.
- Designed and developed an educational chemistry mobile app using React Native.

PROJECTS

Voxel-Entity Game Engine **2025 - Present**

- Developing a 3D voxel rendering game engine to simplify development of 3D cube based games.
- Implemented GPU interaction with Java through JNI and wrote Metal and GLSL shaders both implementing the same rendering algorithm.
- Implemented a .vox file parser to convert 3D MagicaVoxel models into Java objects for physics and rendering.
- Designed JSON serialization utilities enabling networking and data handling across systems.
- Implemented a Bounding Volume Hierarchy (BVH) rendering algorithm for efficient rendering of entities that can exist in floating point space and have dynamic voxel sizes.

Slyde Programming Language **2025 - Present**

- Designed and implemented a compiled programming language using Java.
- Used ANTLR, a language parsing tool, to build a grammar parser that extracts specific patterns from text and encodes these patterns into an abstract syntax tree (AST) that defines the programming language.
- Implemented a compilation pipeline using LLVM IR as an intermediary, allowing programs to be compiled with Clang into machine code which can be directly executed.

SKILLS AND INTERESTS

- Programming Languages: React Native, Java, Javascript, Python (basic), C++(basic), Metal(basic), GLSL(basic)
- Tools and Technologies: Git, ANTLR, LLVM build tools, JNI, JSON, MagicaVoxel